

INTERNATIONAL GCSE **CHEMISTRY**

9202/1

Paper 1

Mark scheme

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Version: 1.0 Final



2 3 B Y 9 2 0 2 / 1 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

- 2.1** In a list of acceptable answers where more than one mark is available 'any **two** from' is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.
- 2.2** A bold **and** is used to indicate that both parts of the answer are required to award the mark.
- 2.3** Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.
- 2.4** Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that 'right + wrong = wrong'.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	Response	Marks awarded
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols / formulae

If a student writes a chemical symbol / formula instead of a required chemical name, full credit can be given if the symbol / formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(...) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

3.9 Ignore

'Ignore' is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do not accept

'Do **not** accept' means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.1	27		1	AO2 3.12h

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.2	13		1	AO2 3.1.2f/h

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.3	+1		1	AO1 3.1.2d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.4	(element)		1	AO1 × 1 AO2 × 1 3.1.3a/b
	any one from: • boron • gallium • indium • thallium • nihonium (reason) same number of electrons in outer shell	allow B allow Ga allow In allow Tl allow Nh allow in the same group MP2 is dependent on MP1 being awarded	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.5	aluminium + chlorine $\rightarrow$ aluminium chloride	allow Al for aluminium allow Cl ₂ for chlorine allow AlCl ₃ for aluminium chloride	1	AO2 3.6.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.6	54 (g)		1	AO2 3.6.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.7	(relative formula mass of AlO ₄ = 27 + (16×4)=) 91	allow correctly determined element from an incorrect relative atomic mass	1	AO2 3.6.2a
	(relative atomic mass of X = 122 – 91=) 31		1	
	phosphorous / P		1	

Total Question 1		10
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Question 2

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.1	hydrocarbons		1	AO1 3.10.1.1b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.2	propane		1	AO1 3.10.1.2b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.3	C_nH_{2n+2}		1	AO1 3.10.1.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.4	crude oil is heated		1	AO1 3.10.1.1c
	to vaporise (the hydrocarbons / alkanes)	allow to evaporate / boil (the hydrocarbons / alkanes)	1	
	there is a temperature gradient in the column	allow the column is hotter at the bottom	1	
	vapours / fractions / hydrocarbons / alkanes condense at their boiling points or vapours / fractions / hydrocarbons / alkanes condense at different levels in the column	allow vapours / fractions / hydrocarbons / alkanes with the highest boiling points collect at the bottom of the column	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.5	low flammability	allow difficult to ignite	1	AO3 3.10.1.2c
	viscous	allow does not flow (easily)	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.6	15.4 (%)		1	AO2 3.10.1.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.7	$(\text{mass}) = \frac{20.5}{100} \times 143$ $= 29.315$ $= 29.3 \text{ (kg)}$	allow an answer correctly rounded to 3 significant figures from a calculation using the values in the question and Table 1	1 1 1	AO2 3.10.1.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.8	(moles of gaseous product = $3 + 5 + 9 =$) 17 (moles of gas produced = $17 \times 0.75 =$) 12.75 alternative approach $(0.75 \times 3) + (0.75 \times 5) + (0.75 \times 9)$ (1) = 12.75 (1)		1 1	AO2 3.10.1.2d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.9	insufficient oxygen (for complete combustion)		1	AO1 3.10.1.2d

Total Question 2		16
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Question 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.1	to clean the (metal) wire or to remove any other substances	allow to clean the (metal) loop	1	AO4 3.4.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.2	crimson		1	AO1 3.4.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.3	allow both contain lithium (ions)	allow both contain Li ⁺ (ions)	1	AO2 3.4.3a 3.3.1.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.4	(test) add (dilute) acid		1	AO1 3.4.3d
	(result) (bubbles of) carbon dioxide produced	allow bubble gas produced through lime water which turns cloudy white MP2 is dependent on MP1 being awarded	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.5	(number of moles of $\text{Li}_2\text{CO}_3 =$) $\frac{3.70}{74}$	allow correct use of an incorrectly determined value for moles of lithium carbonate	1	AO2 3.6.3a
	$= 0.05$		1	
	(number of moles of lithium ions $= 0.05 \times 2) = 0.1$		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.6	(lithium) iodide		1	AO3 3.4.3e 3.5.2c
	the yellow precipitate is silver iodide or this is the test for iodide ions		1	

Total Question 3		10
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Question 4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.1	alkali metals		1	AO1 3.7.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.2	are soft and easily cut		1	AO1 3.7.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.3	all points correctly plotted	allow a tolerance of $\pm \frac{1}{2}$ a small square allow 1 mark for 3 points correct ignore any attempt at a line of best fit	2	AO2 3.7.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.4	63 (°C)	allow a value in the range 53 to 73 (°C) allow a value determined from the line of best fit	1	AO3 3.7.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.5	any one from: • bubbles • (sodium) moves around • (sodium) becomes smaller • (sodium) melts into a ball	allow fizzing allow sodium disappears	1	AO3 3.7.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.6	(rubidium) is more dense than water		1	AO1 3.7.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.7	$2\text{K} + 2\text{H}_2\text{O} \rightarrow 2\text{KOH} + \text{H}_2$	allow multiples allow 1 mark for KOH allow 1 mark for H_2	3	AO2 3.6.1a 3.7.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.8	(unit conversion) $0.600 \text{ dm}^3 = 600 \text{ cm}^3$		1	AO2 3.6.3a 3.6.4a
	(moles = $\frac{45}{600} \times 0.3 =$) 0.0225	allow correct use of incorrect / no unit conversion	1	
	mass = 0.0225×56	allow correct use of incorrect value for number of moles	1	
	= 1.26 (g)		1	
	alternative method			
	(concentration of KOH =) $\frac{0.3}{0.6} = 0.5 \text{ (mol/dm}^3\text{) (1)}$			
	(unit conversion) $45.0 \text{ cm}^3 = 0.045 \text{ dm}^3 \text{ (1)}$			
	(number of moles in $45.0 \text{ cm}^3 = 0.5 \times 0.045 =$) 0.0225 (1)	allow correct use of incorrect / no unit conversion		
		allow correct use of incorrectly determined concentration		
	(mass = $0.0225 \times 56 =$) 1.26 (g) (1)	allow correct use of incorrect value for number of moles		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.9	reactivity increases (going down the group)	allow energy level for shell	1	AO1 3.7.1b/f
	(because) the outer electron / shell is further from the nucleus	allow (because) there are more shells	1	
		allow (because) the atoms get larger		
	(so) there is less (electrostatic) attraction between the nucleus and the outer electron / shell	allow (so) there is more shielding between the outer electron and the nucleus	1	
	(so) the atom loses an electron more easily	allow (so) less energy is needed to remove the (outer) electron	1	

Total Question 4		18
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Question 5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.1	1		1	AO1 3.5.1f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.2	HNO ₃		1	AO1 3.5.1c 3.6.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.3	(aq)		1	AO1 3.6.1b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.4	ZnSO ₄		1	AO2 3.2.1f

Question	Answers	Mark	AO/ Spec. Ref.
05.5	Level 3: The method would lead to the production of a valid outcome. The key steps are identified and logically sequenced.	5–6	AO4 3.5.2a/b
	Level 2: The method would not necessarily lead to a valid outcome. Most steps are identified, but the plan is not fully logically sequenced.	3–4	
	Level 1: The method would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.	1–2	
	No relevant content	0	
	Indicative content: • use zinc oxide and sulfuric acid • add zinc oxide to the (sulfuric) acid • in a beaker • stir • continue adding until the zinc oxide is in excess • shown by excess solid • filter (the reaction mixture) • to remove the excess zinc oxide • heat the solution • using a water bath or electric heater • to crystallisation point • leave the solution to crystallise • pat crystals dry with filter paper		

Total Question 5	10
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Question 6

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.1	any one from: - concentration (of electrolyte / solution) - volume (of electrolyte / solution) - temperature (of electrolyte / solution)		1	AO4 3.9.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.2	(metal) nickel (reason) (nickel) did not produce a potential difference	allow (nickel) did not react	1 1	AO3 3.9.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.3	(most reactive) magnesium vanadium cobalt nickel (least reactive) copper (reason) the higher the potential difference the more reactive the metal	allow 1 mark for (most reactive) magnesium vanadium cobalt allow 1 mark for nickel (least reactive) copper allow the higher the voltage the more reactive the metal	2 1	AO3 3.9.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.4	(electrode X) copper and (electrode Y) vanadium	allow 1 mark for electrode X = vanadium and electrode Y = copper	2	AO3 3.9.3a
	as $0.90 - -0.59 = 1.49\text{V}$		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.5	$\text{Al} \rightarrow \text{Al}^{3+} + 3\text{e}^-$	allow 1 mark for $\text{Al} \rightarrow \text{Al}^{3+} + \text{e}^-$ with no / incorrect balancing numbers	2	AO2 3.3.2f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.6	the reactants are used up		1	AO1 3.9.3a

Total Question 6		12
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Question 7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.1	(metals have) delocalised electrons		1	AO1 3.2.2g
	(the delocalised electrons) move through the metal / structure		1	
	(so the delocalised electrons) transfer energy		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.2	(because) layers (of atoms / ions)		1	AO1 3.3.1a
	slide over each other	allow atoms / ions can slide over each other	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.3	magnesium atom loses two electrons	allow 1 mark for magnesium loses and fluorine gains electron(s)	1	AO2 3.2.1c
	fluorine atoms gain one electron		1	
	two fluorine atoms react with one magnesium atom		1	
	to form ions that have the electron arrangement of a noble gas	allow to gain full outer shells of electrons allow to form Mg ²⁺ ions and (two) F ⁻ ions	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.4	giant structure (of ions)	allow lattice	1	AO1 3.2.2a
	(with) strong electrostatic forces of attraction between oppositely charged ions		1	
	in all directions		1	
	(so) a large amount of energy is needed to overcome (the forces of attraction)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.5	oxide ions have a higher (negative) charge than fluoride ions or (electrostatic) forces of attraction in magnesium oxide are stronger		1	AO3 3.2.2a

Total Question 7		14
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